L7: The Chinese Remainder Theorem

Example:- Find a positive integer having a remainder of 2 when divided by 3, a remainder of 1 when divided by 4 , and a remainder of 3 when divided by 5 .
In other words, we want to find a positive integer x such that:

$$\begin{aligned} x &\equiv 2 (\bmod 3) \\ x &\equiv 1 (\bmod 4) \\ x &\equiv 3 (\bmod 5) \end{aligned}$$

This is a system of linear congruences in one variable.
Theorem 2.9 (Chinese Remainder Theorem):
Let $m_1, m_2, \ldots, m_n$ be pairwise velatively prime positive integers and let $b_1, \ldots, b_n$ be any integers. Then the system of linear congruences in one variable given by

$$\begin{aligned} x &\equiv b_1 \,(\bmod m_1) \\ x &\equiv b_2 \,(\bmod m_2) \\ \\ x &\equiv b_n \,(\bmod m_n) \end{aligned}$$

has a unique solution modulo $m_1 m_2 \ldots m_n$.
Proof: We first construct a solution. Let

$$\begin{aligned} &M := m_1 m_2 \ldots m_n, \text{ and} \\ &M_i := M_1 m_i \text{ for } i = 1, 2, \ldots, n. \end{aligned}$$

Thus $(M_i, m_i) = 1$ for each i, and hence $m_i x_i \equiv 1 \,(\bmod m_i)$ has a solution for each i by Corollary 2.8. Consider

$$x = b_1 M_1 x_1 + b_2 M_2 x_2 + \ldots + b_n M_n x_n.$$

This is a solution to the desired system since for each $i = 1, 2, \ldots, n$ we have

$$\begin{aligned} x &= b_1 M_1 x_1 + \cdots + b_i M_i x_i + \cdots + b_n M_n x_n \\ &\equiv 0 + \cdots + b_i + \cdots + 0 \,(\bmod m_i) \\ &\equiv b_i \,(\bmod m_i) \,. \end{aligned}$$

It remains to show uniqueness of the solution Modulo M. Let x' be another solution to the given system of linear congruences. Then we have that $x' \equiv b_i \,(\bmod m_i)$ for all i.

Since $x \equiv b_i \,(\bmod m_i)$, we have that $x \equiv x' \,(\bmod\ m_i\ )$ for all i, or equivalently $\rightarrow m_i \mid x - x'$ for all i. Thus $M \mid x - x'$ since $m_1, m_2, \ldots, m_n$ are pairwise velatively prime. Thus $x \equiv x' (\bmod M)$, and the proof is complete.

Example (continued): Recall that we needed to Solve:

$$\begin{aligned} x &\equiv 2 (\bmod 3) \\ x &\equiv 1 (\bmod 4) \\ x &\equiv 3 (\bmod 5) \end{aligned}$$

Using the notation of Theorem 2.9 we have

$$\begin{aligned}
M &= 3 \cdot 4 \cdot 5 = 60 \\
M_1 &= M/M_1 = 60/3 = 20 \\
M_2 &= M/m_2 = 60/4 = 15 \\
M_3 &= M/m_3 = 60/5 = 12.
\end{aligned}$$

We now solve the linear congruences $M_i x_i \equiv 1 \pmod{m_i}$ for $i = 1, 2, 3$. In general, one would use the Euclidean algor ithm to
to solve these congrnences. We solve each one by inspection (apribsi knowing that
$M_i x_i \equiv 1 \pmod{m_i}$ has a unique solution by
Conollary 2.8). Observe that $M_1 x_1 \equiv 1 \pmod{m_1}$
becomes $20x_1 \equiv 1(\mathrm{mod} 3)$
iff $\quad 2x_1 \equiv 1(\mathrm{mod} 3)$
iff $\quad x_1 \equiv 2(\mathrm{mod} 3)$.
Observe that $M_2 x_2 \equiv 1 \pmod{m_2}$ becomes
iff
$15x_2 \equiv 1(\mathrm{mod} 4)$
$3x_2 \equiv 1(\mathrm{mod} 4)$
iff
$x_2 \equiv 3(\mathrm{mod} 4)$
Finally, observe that $M_3 x_3 \pmod{M_3}$ becomes
iff
$12x_3 \equiv 1(\mathrm{mod} 5)$
$2x_3 \equiv 1(\mathrm{mod} 5)$
ift $\quad x_3 \equiv 3(\mathrm{mod} 5)$.
Thus we put $x_1 = 2, x_2 = 3, x_3 = 3$. The desired solution is

$$\begin{aligned}
x &= b_1 M_1 x_1 + b_2 M_2 x_2 + b_3 M_3 x_3 \\
&= (2)(20)(2) + (1)(15)(3) + (3)(12)(3) \\
&= 233 \\
&\equiv 53(\mathrm{mod} 60).
\end{aligned}$$

The minimum positive solution to the System is 53.

Wilson's Theorem

Lemma 2.10: Let p be a prime and let $a \in \mathbb{Z}$. Then a is its own inverse modulo p if and only if $a \equiv \pm 1(\mathrm{mod} p)$.
Proof: ($\Rightarrow$) Assume that a is its own inverse modulo p. Then $a^2 \equiv 1(\mathrm{mod} p)$. Thus $p \mid a^2 - 1$, or equivalently, $p((a-1)(a+1)$. By Lemmal. 14 we have that $p \mid a - 1$ or $p \mid a + 1$. Thus $a \equiv \pm 1(\mathrm{mod} p)$, as desired.
($\Leftrightarrow$) Assume that $a \equiv \pm 1(\mathrm{mod} p)$. Then $a^2 \equiv 1(\mathrm{mod} p)$, and a is its own inverse modulo p, as desired.

Theorem 2.11 (Wilson): Let p be a prime. (6) Then $(p-1)! \equiv -1(\bmod p)$.

- John wilson conjectured the result based on numerical evidence, but could not prove it.
- The Theorem was first proved by Lagrange in 1771.
  Example (idea behind the proof): We prove that $(11-1)! \equiv -1(\bmod 11)$. We have

$$10! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$$

By Lemma 2.10 , the only integers between 1 and 10 inclusive that are their own inverses modulo II are I and 10. We now take each integer from 2 to 9 inclusive and pair it with its inverse modulo II:

$$\begin{aligned} 2 &\leftrightarrow 6 \\ 3 &\leftrightarrow 4 \\ 5 &\leftrightarrow 9 \\ 7 &\leftrightarrow 8 \end{aligned}$$

The product of each pair is congruent to
I modulo II, from which

$$\begin{aligned} 10! &= 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 \\ &= 1 \cdot (2 \cdot 6)(3 \cdot 4)(5 \cdot 9)(7 \cdot 8) \cdot 10 \\ &\equiv 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot (-1)(\bmod 11) \\ &\equiv -1(\bmod 11). \end{aligned}$$

as desired.

Proof (of Wilson's Theorem): The statement of the Theorem is easily checked for $p=2$ and $p=3$. Assume that $p>3$. Each $a \in \mathbb{Z}$ with $1 \leqslant a \leqslant p-1$ has a unique inverse modulo p by Corollary 2.8. Without loss of generality, we assume that the inverse of such an a, say a', satisfies $1 \leqslant a' \leqslant p-1$. Now, by Lemma 2.10, the only integers that are their own inverses modulo p are those integers congruent to $\pm 1(\bmod p)$, so the only integers a with $1 \leq a \leq p-1$ that are their own inverses modulo p are 1 and $p-I$. Therefore
for each integer a with $2 \leqslant a \leqslant p-2$, there exists a unique different integer a' with $2 \leqslant a' \leqslant p-2$ such that $aa' \equiv 1(\bmod p)$. Thus we can group the numbers $2,3,\ldots,p-2$, into pairs a, a', such that

$$aa' \equiv 1(\bmod p). \quad (*)$$

The product of the left sides of the $\frac{p-3}{2}$ congruences $(*)$ is $(p-2)$! after rearrangement. The right sides multiply to 1. Thus

$$(p-2)! \equiv 1(\bmod p) \cdot (**)$$

Multiplying both sides of (*) by $p-1$ we obtain

$$(p-2)!(p-1) \equiv p-1(\bmod p),$$

and thus

$$(p-1)! \equiv -1(\bmod p),$$

as desired.

- The converse of Wilson's Theorem is also true.

Proposition 2.12: Let $n \in \mathbb{Z}$ with $n > 1$. If $(n-1)! \equiv -1(\bmod n)$, then n is a prime. Proof: Let $n = ab$ where $a, b \in \mathbb{Z}$ with $1 \leqslant a < n$. It suffices to show that $a = 1$, so that any such factorisation of n is trivial (consisting of 1 and n only). Since $1 \leqslant a < n$, we have that a $\mid (n-1)!$. Now the hypothesis $(n-1)! \equiv -1(\bmod n)$ implies that $n \mid (n-1)! + 1$. Since $a \mid n$ we have that a $\mid (n-1)! + \pm$ by Proposition 1.1. Thus a$\mid$ $(n-1)! + 1 - (n-1)! = 1$ by Proposition 1.2 and thus $a \mid 1$. So $a = 1$, as required.

Remark: •Proposition 2.12 gives a primality test! Although, it is very ineff.cet. inefficient i.e. Computation of $(n-1)$! requires at least $n-3$ multiplications modulo n)

- wilson's name is also cessociated to a
  class of primes.

Definition: A prime p is said to be a wilson prime if $(p-1)! \equiv -1 \left(\bmod p^2\right)$.

- Only three wilson primes are known:
  $5, 13, 563$. It is not known whether
  there are finitely or infinitely many wilson primes.

Fermat's Little Theorem

Theorem 2.13 (Fermat's Little Theorem): Let p be a prime and $a \in \mathbb{Z}$. If $p + a$, then

$$a^{p-1} \equiv 1(\bmod p).$$

Proof: Consider the $p-1$ integers given by $a, 2a, \ldots, (p-1)a$. Note that $p + ia$ for $i = 1, 2, \ldots, p-1$ (if plia for some i, then pla or pli by Lemma 1.14, which is impossible). Note also that no two of the given $p-1$ integers are congruent modulo P. Why? Since $p + a$, the inverse of $a_,$
say a', exists by Corollary 2.8 , so if (11)
$ia \equiv ja(\bmod p)$ with $i \neq j$, then
i $aa' \equiv jaa'(\bmod p)$, from which $i \equiv j(\bmod p)$,
which is impossible. So the least nonnegative resides modulo p of $a, 2a, \ldots,$
$(p-1)a$, taken in some order must be $1, 2, \ldots, p-1$. Then,
$(a)(2a) \cdots ((p-1)a) \equiv 1 \cdot 2 \cdots (p-1)(\bmod p)$, or equivalently,

$$a^{p-1}(p-1)! \equiv (p-1)!(\mathrm{mod} p).$$

By wilson's Theorem, the congruence becomes

$$-a^{p-1} \equiv -1(\mathrm{mod} p),$$

and so $a^{p-1} \equiv 1(\mathrm{mod} p)$.
Remark: Wilson's Theorem is not needed here.
why?